

Lecture 9: The Matrix (continued)

Instructors:

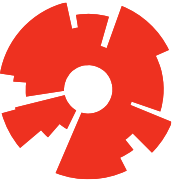
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[Based on slides for „Decoding the Matrix“]



WEEK

FIVE



Agenda

Chapter 4

Lecture 9:

Matrix for 90° Rotation

Matrix for θ° Rotation

Diagonal Matrices

Linear Functions

Linear Functions & Linear Combinations

Kernel

Kernel being one-to-one

Kernel being onto

Matrix-Matrix Multiplication

Matrix-Matrix Multiplication - Transpose





Recap: Some Key Take-aways (Week 4)

A **matrix** has rows and columns, and individual elements are addressed based on their row and column index

Using the transpose of a matrix, one can switch between vector-matrix and matrix-vector multiplication

Matrix-vector multiplication and **vector-matrix multiplication** can be implemented via linear combinations and the dot product

The **null space for a matrix** A ($Null A$) is the vector that multiplied with the matrix gives the zero vector and is a **vector space** (V_1, V_2, V_3)

Link to homogeneous linear systems

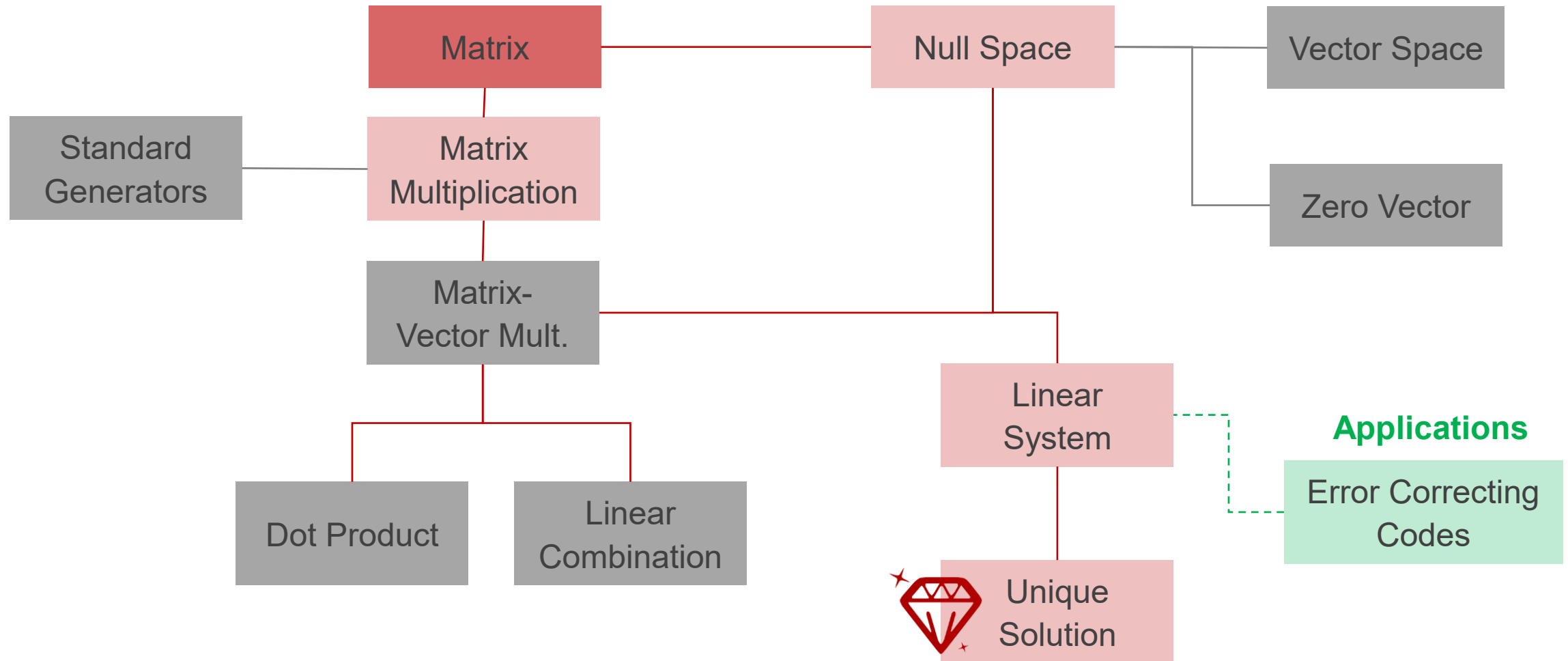
If a vector u is a solution to a linear system (expressed via matrix-vector multiplication: $Ax = b$) then the solution set is $u + Null A$

We previously covered this without matrices



Lecture 8 - Summary

Concepts

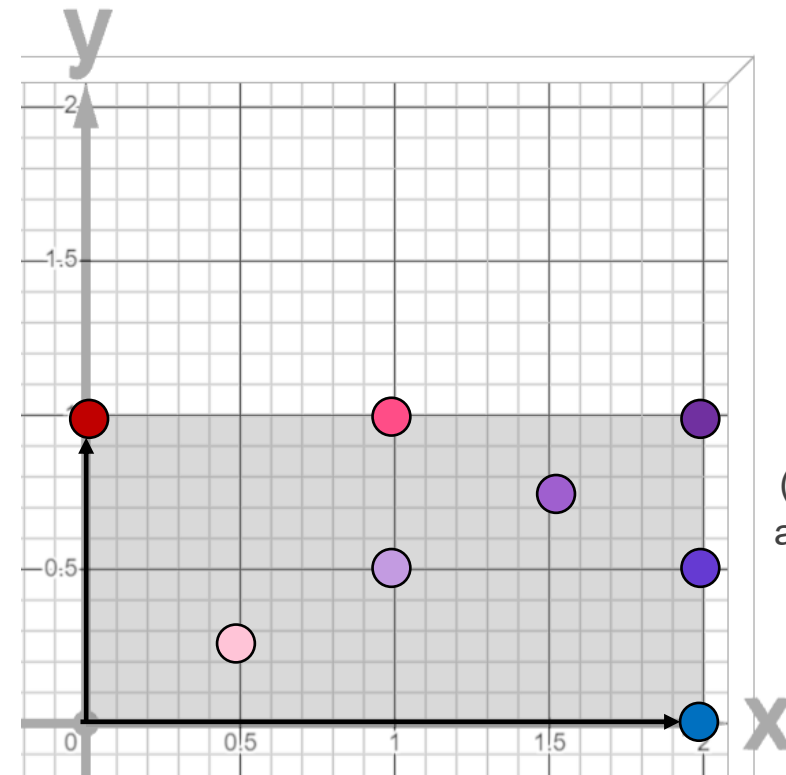
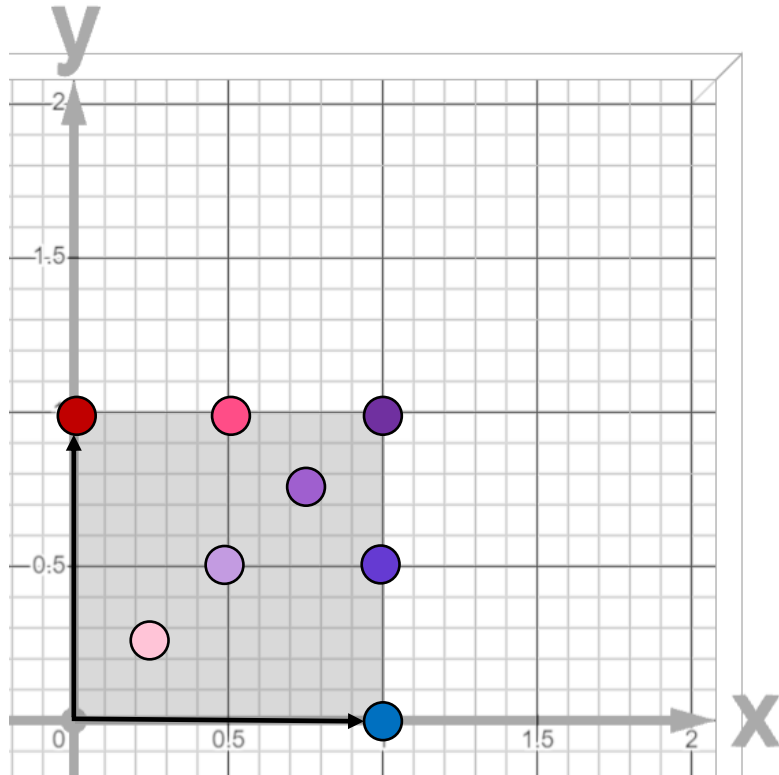


Recap: Matrix Scaling (I)

$$M = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

$$M * e_1 = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} * [1, 0]^T = 1 * \begin{bmatrix} 2 \\ 0 \end{bmatrix} + 0 * \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$M * e_2 = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} * [0, 1]^T = 0 * \begin{bmatrix} 2 \\ 0 \end{bmatrix} + 1 * \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



(We will come back to the area in **grey** (on the right), when we talk about the **determinant**)



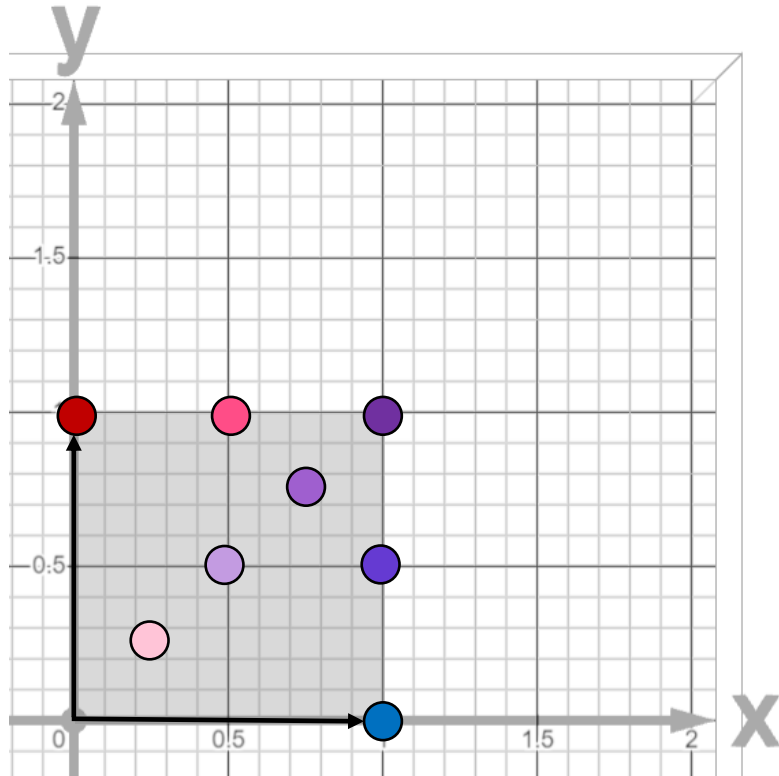


Recap: Matrix Scaling (II)

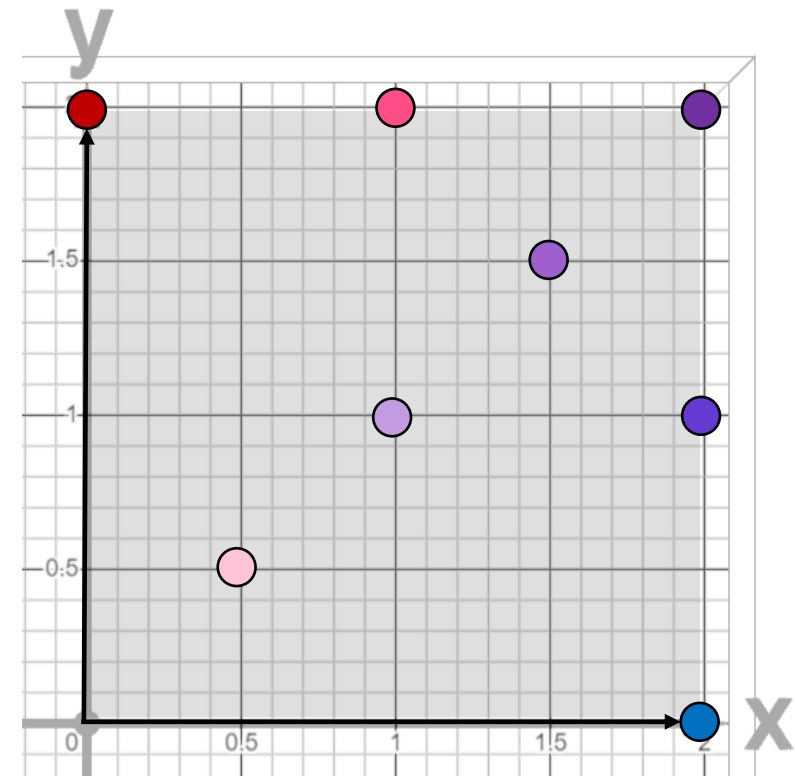
$$M = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$M * e_1 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} * [1, 0]^T = 1 * \begin{bmatrix} 2 \\ 0 \end{bmatrix} + 0 * \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$M * e_2 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} * [0, 1]^T = 0 * \begin{bmatrix} 2 \\ 0 \end{bmatrix} + 1 * \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$



→ Stretched



→ Stretched

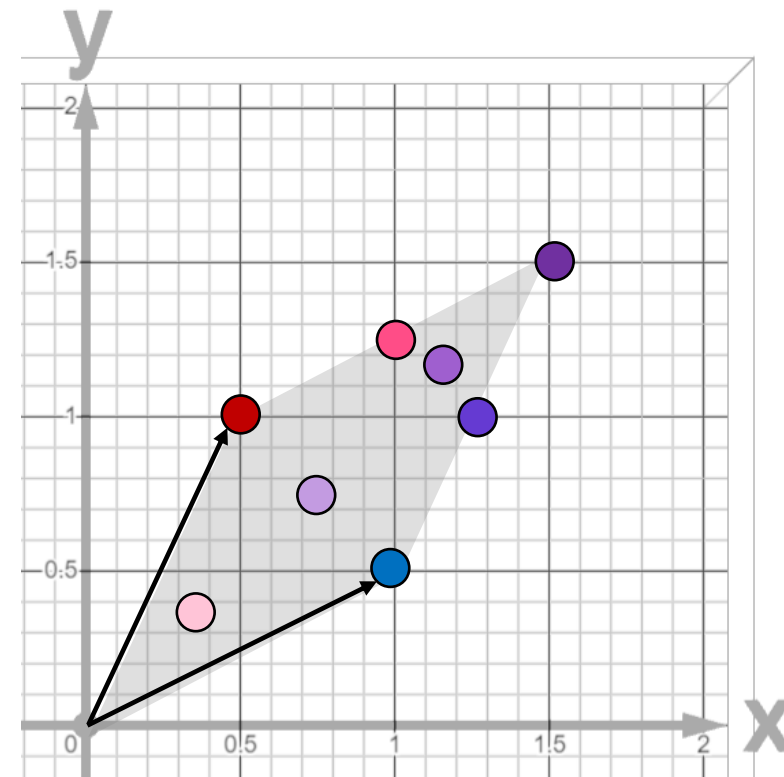
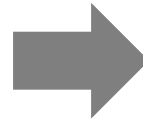
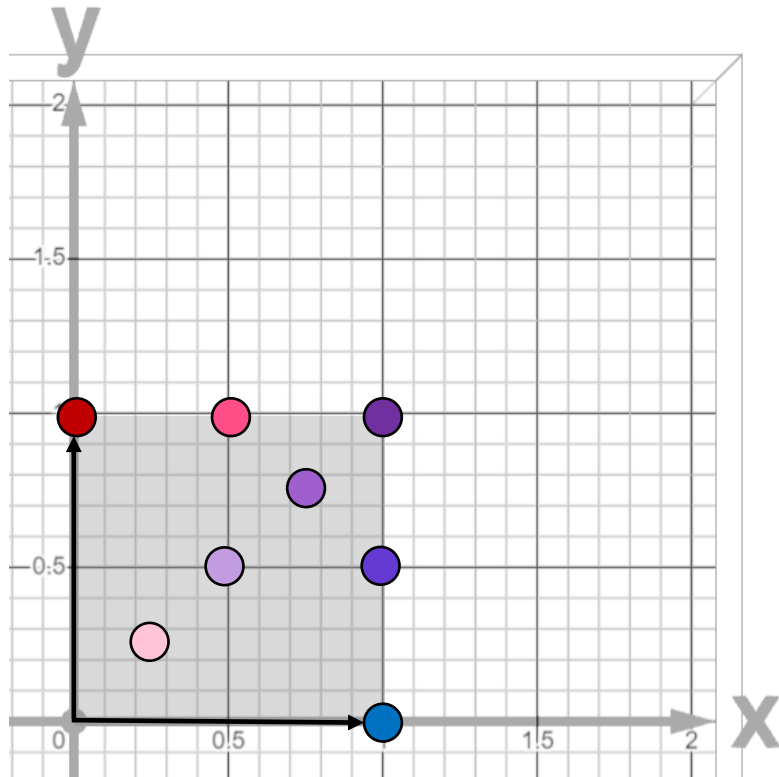


Matrix Transformations (I)

$$M = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}$$

$$M * e_1 = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix} * [1, 0]^T = 1 * \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} + 0 * \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}$$

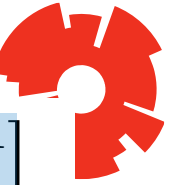
$$M * e_2 = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix} * [0, 1]^T = 0 * \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} + 1 * \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 1 \end{bmatrix}$$



Each point is
still unique
represented

Points in terms of the columns of M

(remember this when we go to coordinate systems and bases later)

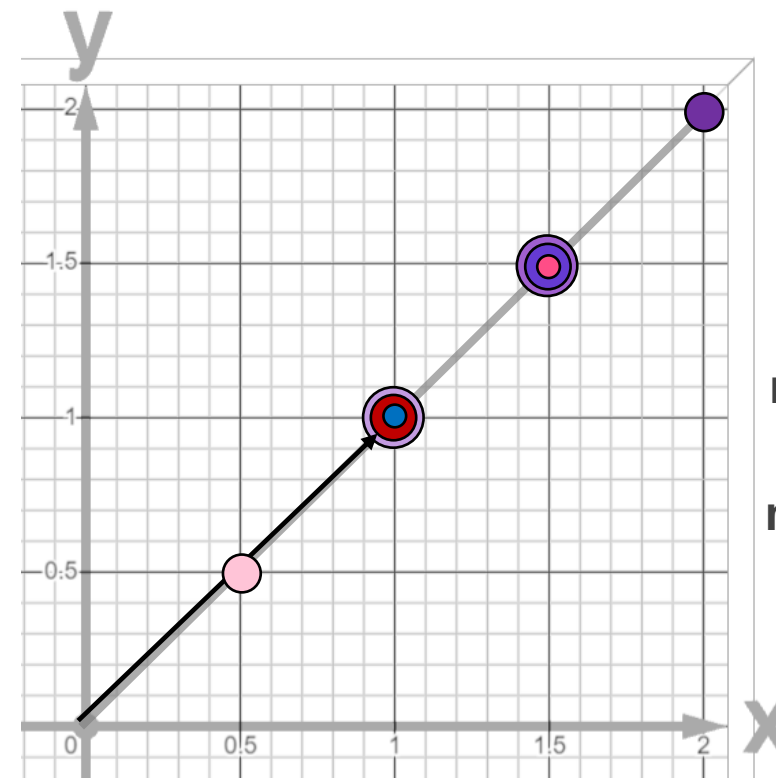
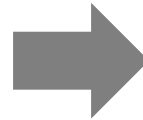
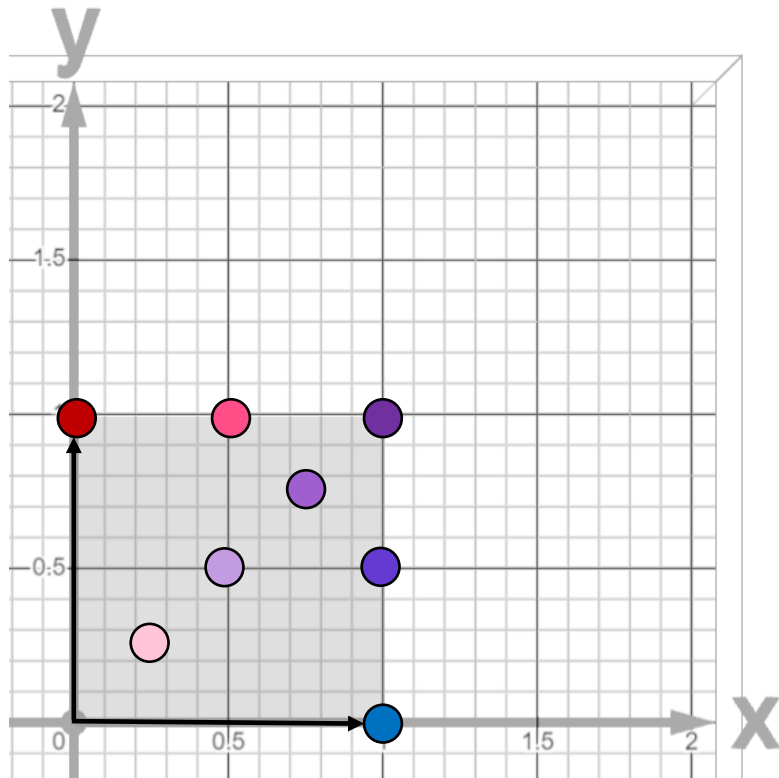


Matrix Transformations (II)

$$M = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad \text{Columns are now the same}$$

$$M * e_1 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} * \begin{bmatrix} 1 \\ 0 \end{bmatrix}^T = 1 * \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 0 * \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$M * e_2 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} * \begin{bmatrix} 0 \\ 1 \end{bmatrix}^T = 0 * \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 1 * \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$



Multiple points
now end up at the
same location –
now only on a line

Points in terms of the columns of M

(remember this when we go to coordinate systems and bases later)

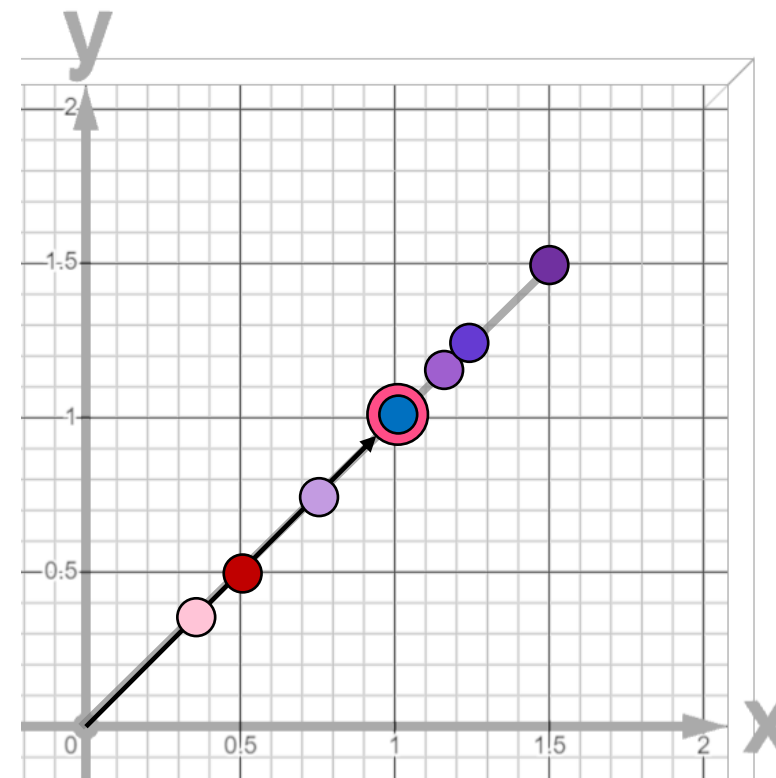
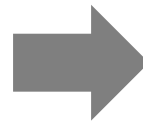
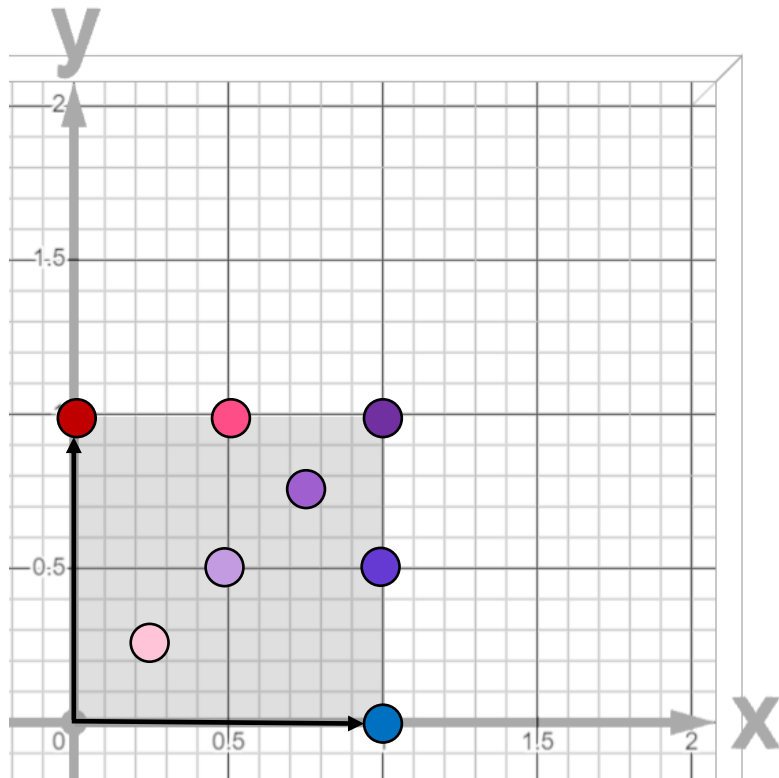


Matrix Transformations (III)

$$M = \begin{bmatrix} 1 & 0.5 \\ 1 & 0.5 \end{bmatrix} \quad \text{Columns are not the same} \\ \text{(but their **span** is!)}$$

$$M * e_1 = \begin{bmatrix} 1 & 0.5 \\ 1 & 0.5 \end{bmatrix} * [1, 0]^T = 1 * \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 0 * \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$M * e_2 = \begin{bmatrix} 1 & 0.5 \\ 1 & 0.5 \end{bmatrix} * [0, 1]^T = 0 * \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 1 * \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$$



Multiple points
end up at the
same location –
only on a line

Points in terms of the columns of M

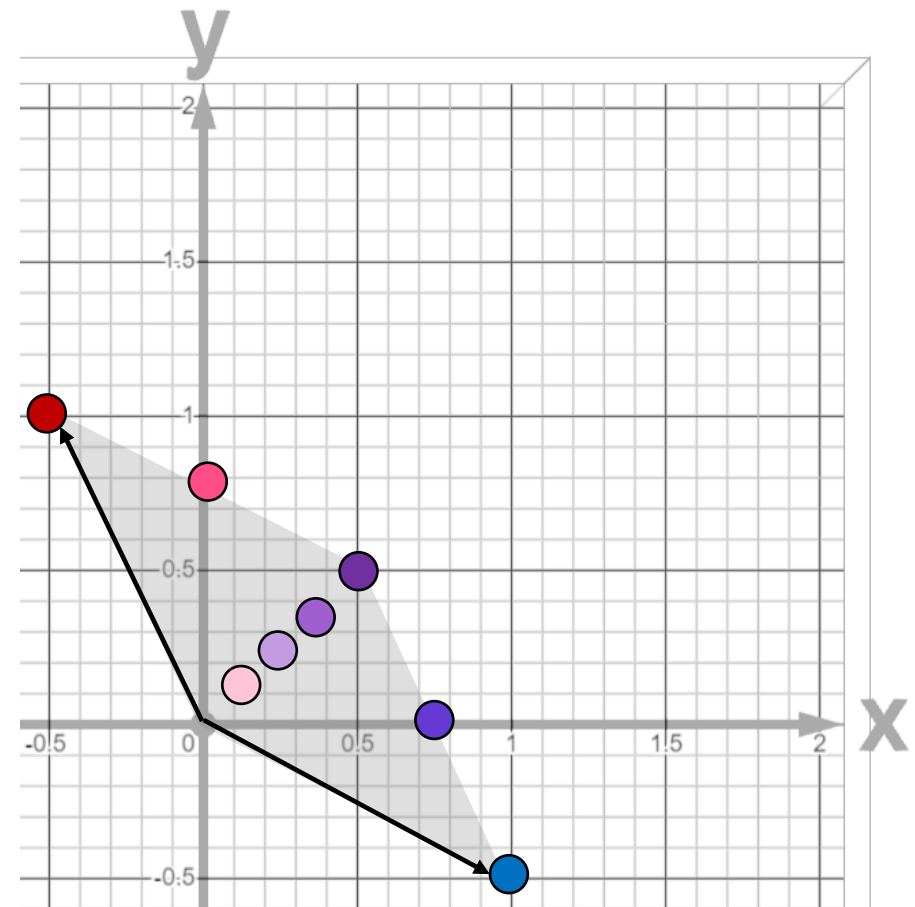
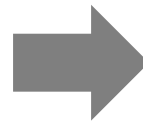
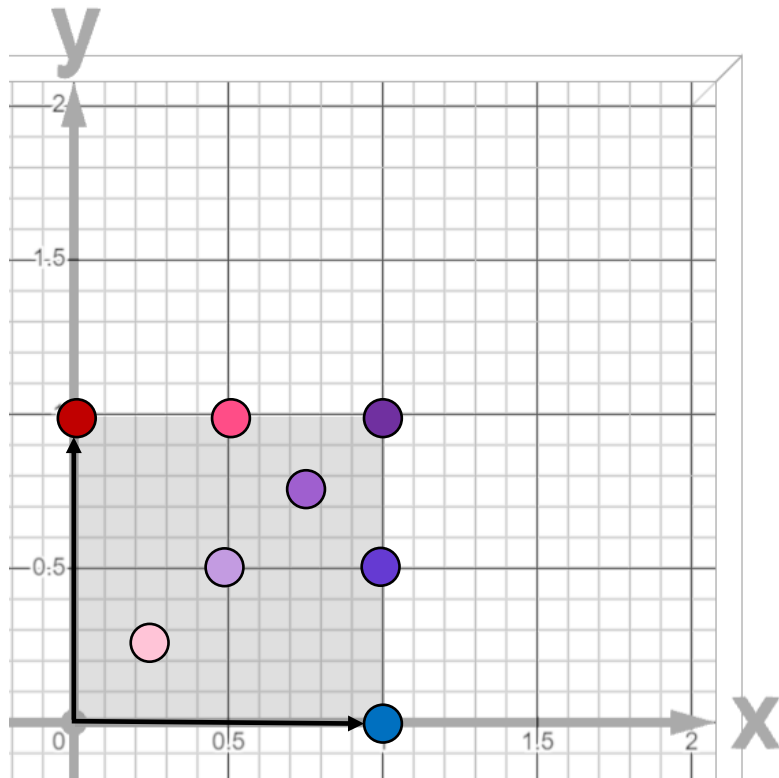
(remember this when we go to coordinate systems and bases later)

Matrix Transformations (IV)

$$M = \begin{bmatrix} 1 & -0.5 \\ -0.5 & 1 \end{bmatrix}$$

$$M * e_1 = \begin{bmatrix} 1 & -0.5 \\ -0.5 & 1 \end{bmatrix} * [1, 0]^T = 1 * \begin{bmatrix} 1 \\ -0.5 \end{bmatrix} + 0 * \begin{bmatrix} -0.5 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -0.5 \end{bmatrix}$$

$$M * e_2 = \begin{bmatrix} 1 & -0.5 \\ -0.5 & 1 \end{bmatrix} * [0, 1]^T = 0 * \begin{bmatrix} 1 \\ -0.5 \end{bmatrix} + 1 * \begin{bmatrix} -0.5 \\ 1 \end{bmatrix} = \begin{bmatrix} -0.5 \\ 1 \end{bmatrix}$$



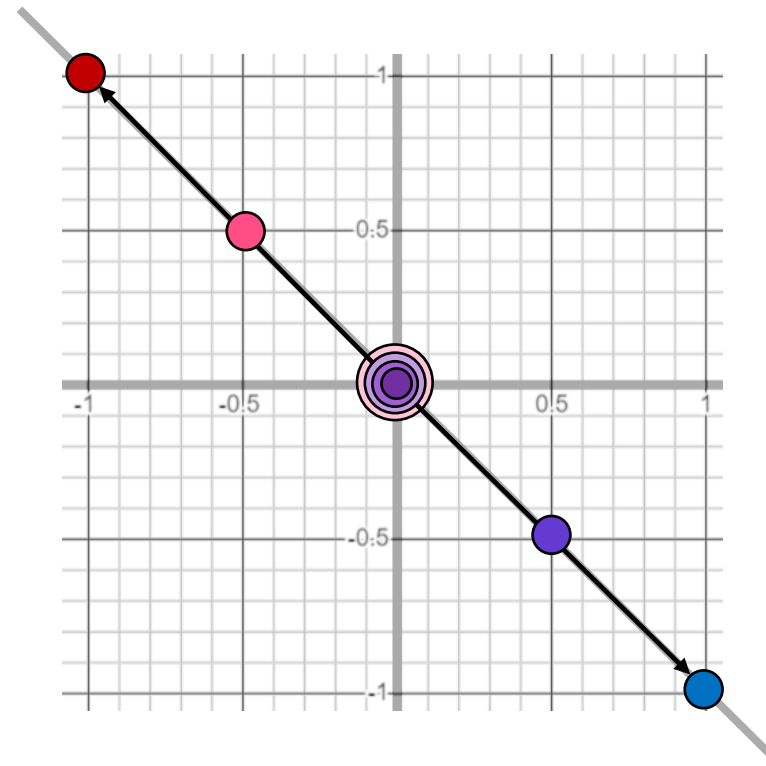
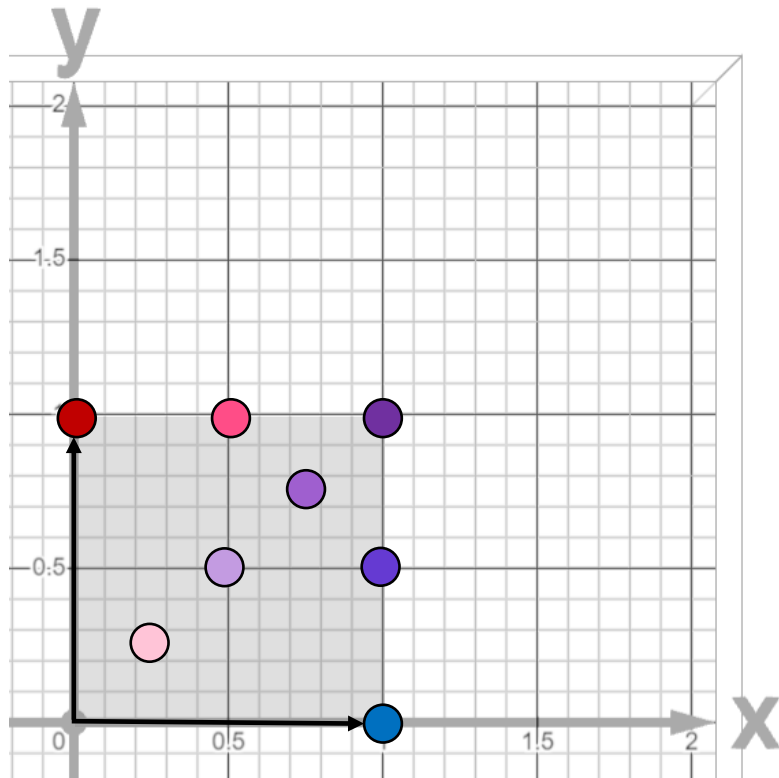


Matrix Transformations (V)

$$M = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \text{ Columns are not the same (but their span is!)}$$

$$M * e_1 = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} * [1, 0]^T = 1 * \begin{bmatrix} 1 \\ -1 \end{bmatrix} + 0 * \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$M * e_2 = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} * [0, 1]^T = 0 * \begin{bmatrix} 1 \\ -1 \end{bmatrix} + 1 * \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$



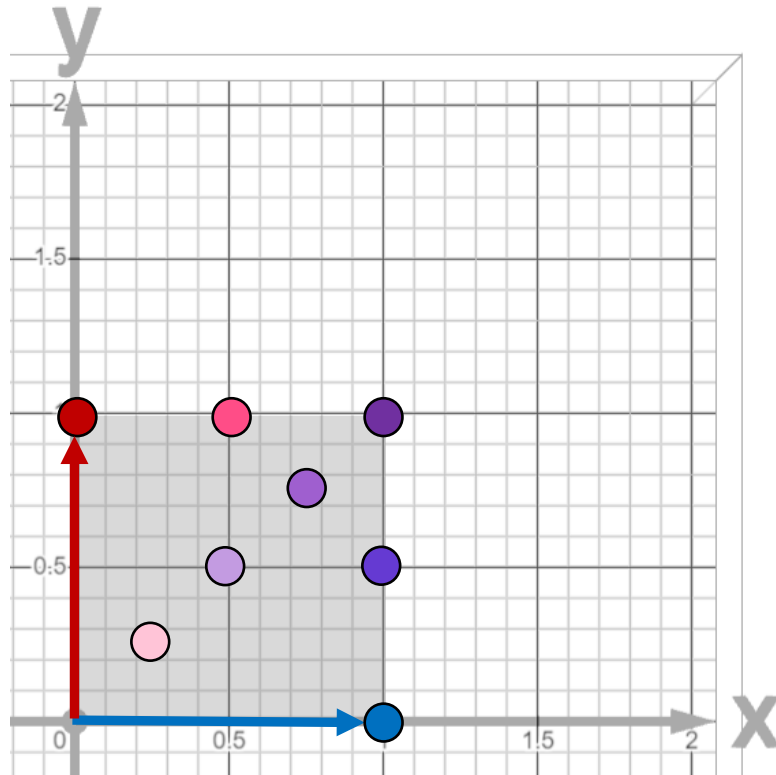
Multiple points
end up at the
same location –
only on a line



Counter-clockwise Matrix - 90° Rotation (I)

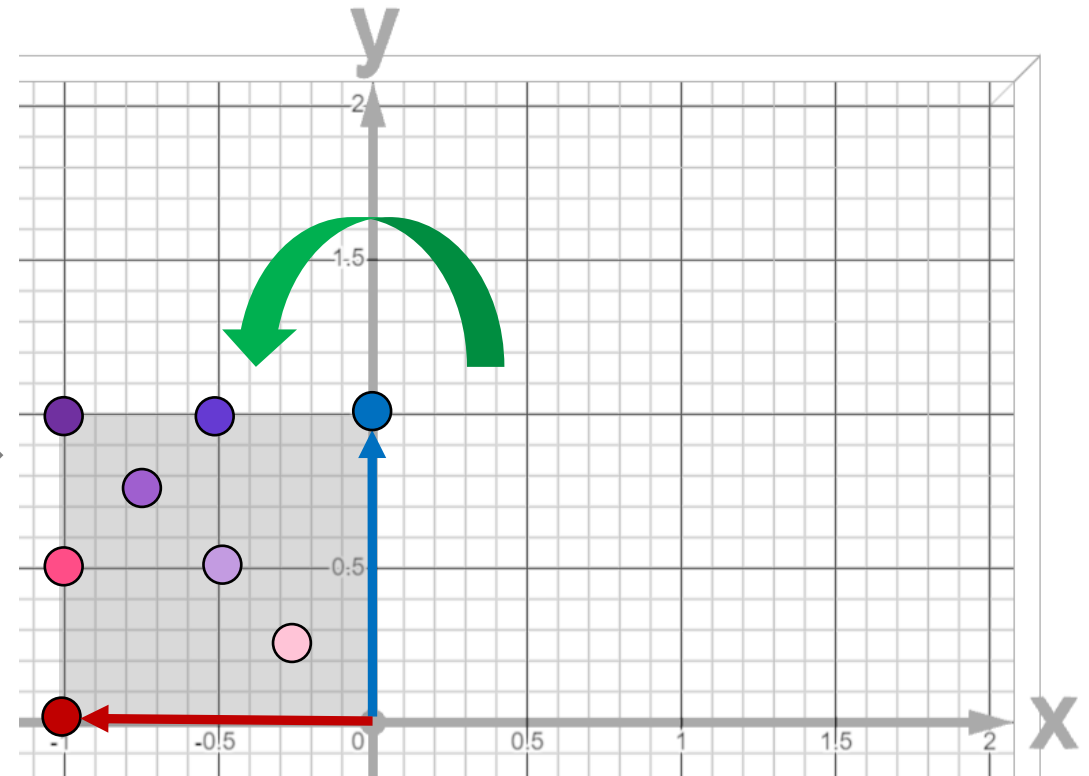
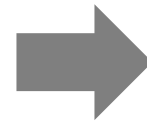
Define $f\left(\begin{bmatrix} x \\ y \end{bmatrix}\right)$ = rotation by 90 degrees

► We know $f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $f\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$



Therefore $M = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

Counter-
clockwise
rotation by 90°





Counter-clockwise Matrix - 90° Rotation (II)

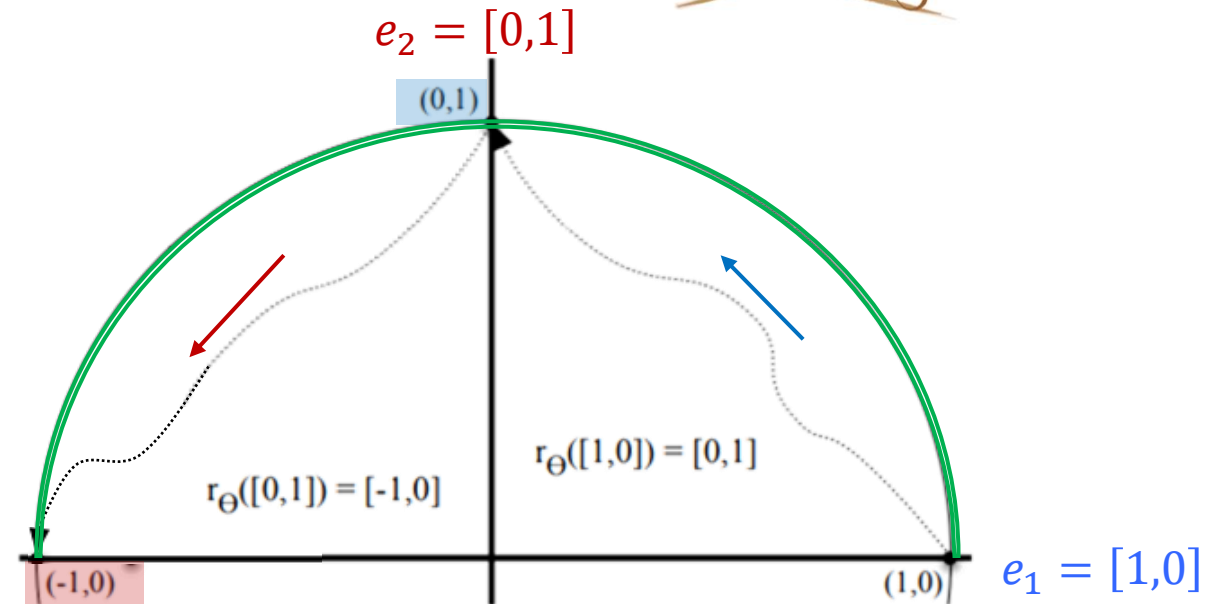
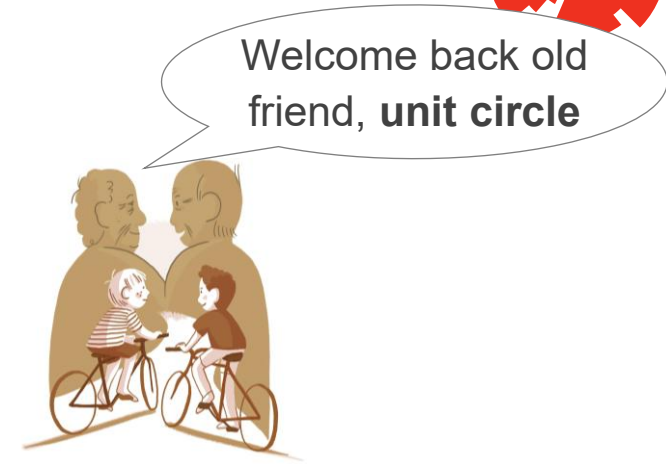
Define $f\left(\begin{bmatrix} x \\ y \end{bmatrix}\right)$ = rotation by 90 degrees

► We know $f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$f\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

Therefore $M = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

As we did with complex numbers (Lecture 2)
We can think of it as moving (the standard generators) along the **unit circle**

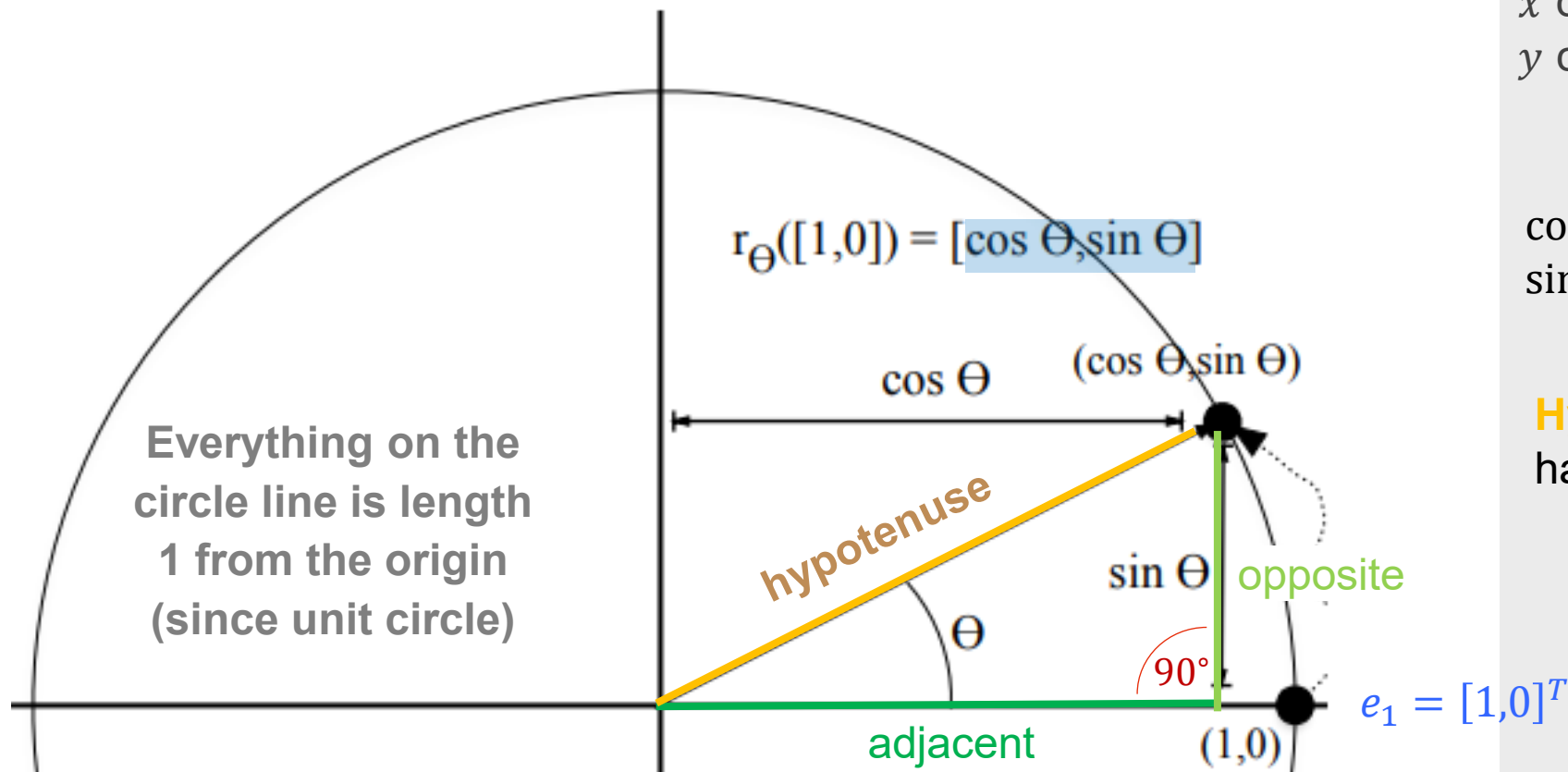




Counter-clockwise

Matrix - θ° Rotation (I)

How to get the coordinates knowing only angle θ
and that we have a unit circle?



For the first column of M (so for e_1)

x coordinate will be the **adjacent**
 y coordinate will be the **opposite**



$\cos \theta = \text{Adjacent} / \text{Hypotenuse}$
 $\sin \theta = \text{Opposite} / \text{Hypotenuse}$



Hypotenuse is on circle line, so
has length **one**



Adjacent = $\cos \theta$

Opposite = $\sin \theta$



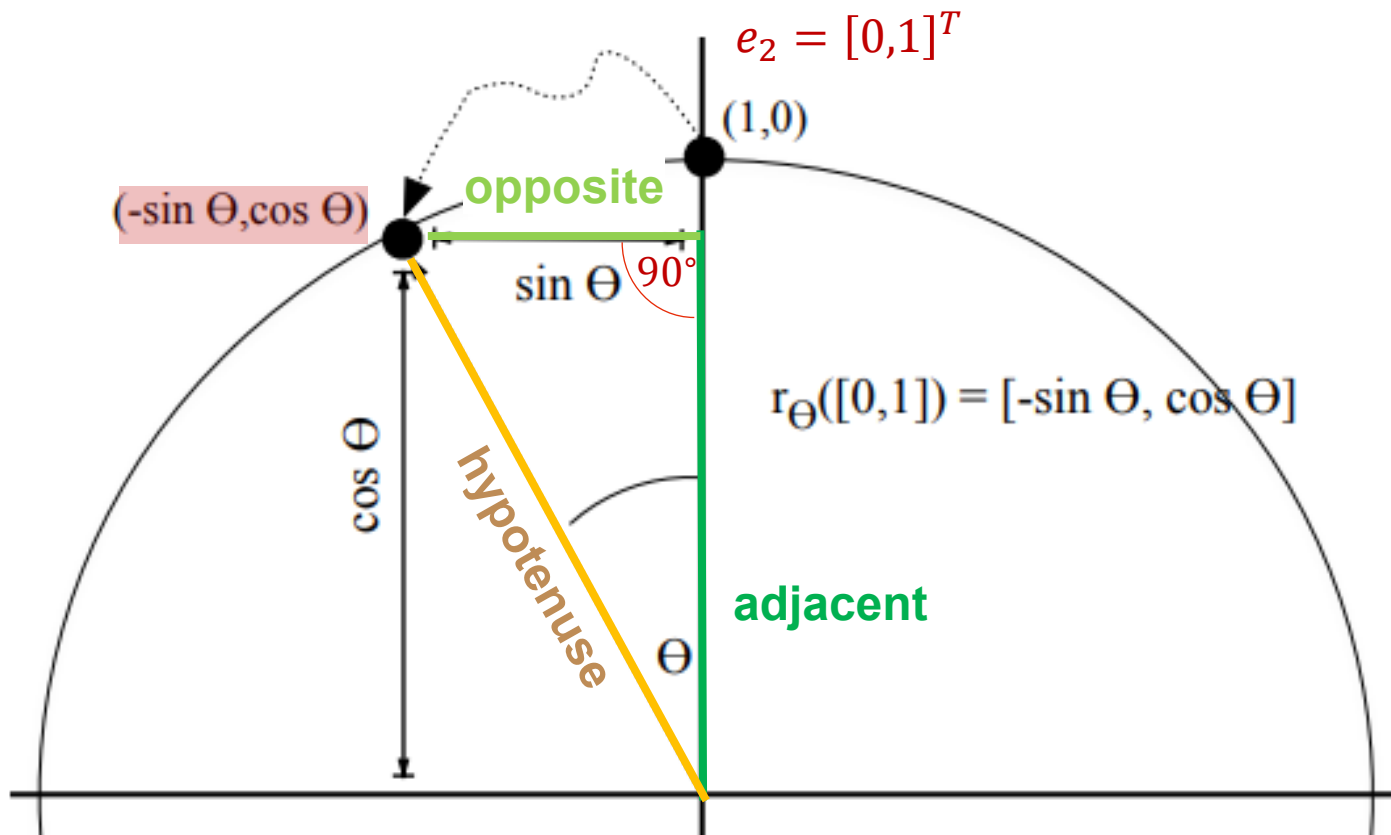
$$[x, y]^T = [\cos \theta, \sin \theta]^T$$



Counter-clockwise

Matrix - θ° Rotation (II)

Same Logic for the second column of M (so for e_2)



x coordinate will be the **-(opposite)**
 y coordinate will be the **adjacent**



$$\cos \theta = \text{Adjacent} / \text{Hypotenuse}$$

$$\sin \theta = \text{Opposite} / \text{Hypotenuse}$$



Hypotenuse is on circle line, so
 has length **one**



⚠ **-(Opposite)** = $-\sin \theta$
Adjacent = $\cos \theta$



$$[x, y]^T = [-\sin \theta, \cos \theta]^T$$



Counter-clockwise Matrix - θ° Rotation (III)

Having done the previous steps, it should come as no surprise that ...

Define $f\left(\begin{bmatrix} x \\ y \end{bmatrix}\right)$ = rotation by θ .

► We know $f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}$ and $f\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix}$ Therefore $M = \left[\begin{array}{c|c} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{array} \right]$

Depending on the (counter-clockwise) rotation angle you want, just plug in the corresponding θ for the **M matrix**



Counter-clockwise

Matrix - θ° Rotation (IV)

$$\text{Therefore } M = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

Let's try some examples

$$\text{Rotation by } 90^\circ \rightarrow \begin{bmatrix} \cos 90^\circ & -\sin 90^\circ \\ \sin 90^\circ & \cos 90^\circ \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$



Looks familiar (see some slides ago)

$$\text{Rotation by } 180^\circ \rightarrow \begin{bmatrix} \cos 180^\circ & -\sin 180^\circ \\ \sin 180^\circ & \cos 180^\circ \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\text{Rotation by } 200^\circ \rightarrow \begin{bmatrix} \cos 200^\circ & -\sin 200^\circ \\ \sin 200^\circ & \cos 200^\circ \end{bmatrix} \approx \begin{bmatrix} -0.94 & 0.34 \\ -0.34 & -0.94 \end{bmatrix}$$



Matrix - Identity Matrix

Consider the function $f : \mathbb{R}^4 \longrightarrow \mathbb{R}^4$ defined by $f(\mathbf{x}) = \mathbf{x}$
This is the identity function on $\mathbb{R}^4$.

Assume $f(\mathbf{x}) = M * \mathbf{x}$ for some matrix M .

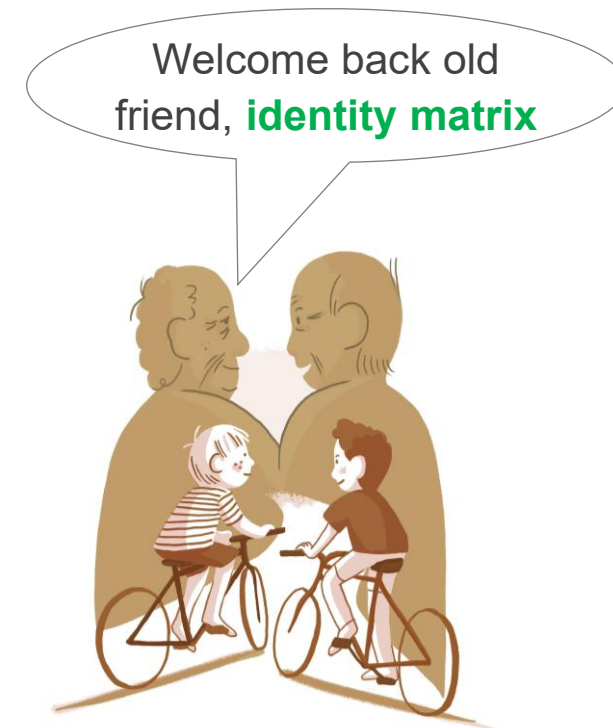
Plug in the standard generators

$\mathbf{e}_1 = [1, 0, 0, 0]$, $\mathbf{e}_2 = [0, 1, 0, 0]$, $\mathbf{e}_3 = [0, 0, 1, 0]$, $\mathbf{e}_4 = [0, 0, 0, 1]$

- ▶ $f(\mathbf{e}_1) = \mathbf{e}_1$ so first column is $\mathbf{e}_1$
- ▶ $f(\mathbf{e}_2) = \mathbf{e}_2$ so second column is $\mathbf{e}_2$
- ▶ $f(\mathbf{e}_3) = \mathbf{e}_3$ so third column is $\mathbf{e}_3$
- ▶ $f(\mathbf{e}_4) = \mathbf{e}_4$ so fourth column is $\mathbf{e}_4$

So $M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Identity function $f(\mathbf{x})$ corresponds to identity matrix $\mathbf{1}$





Matrix - Diagonal Matrix

Let $d_1, \dots, d_n$ be real numbers. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be the function such that $f([x_1, \dots, x_n]) = [d_1x_1, \dots, d_nx_n]$. The matrix corresponding to this function is

$$\begin{bmatrix} d_1 & & \\ & \ddots & \\ & & d_n \end{bmatrix}$$

All elements other
than the diagonal
are zero

Such a matrix is called a diagonal matrix because the only entries allowed to be nonzero form a diagonal.

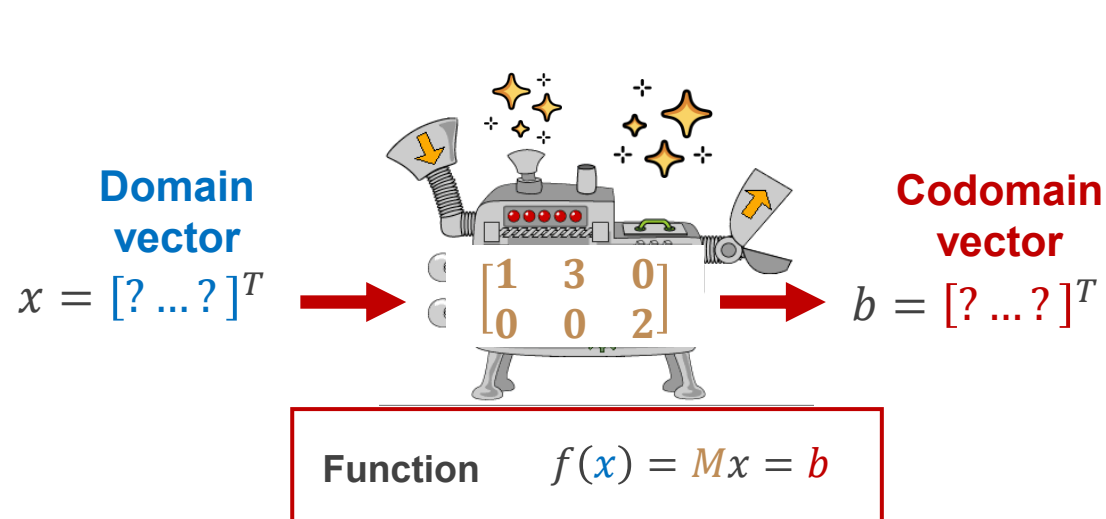
Definition: For a domain D , a $D \times D$ matrix M is a *diagonal* matrix if $M[r, c] = 0$ for every pair $r, c \in D$ such that $r \neq c$.

Special case: $d_1 = \dots = d_n = 1$. In this case, $f(\mathbf{x}) = \mathbf{x}$ (*identity function*)

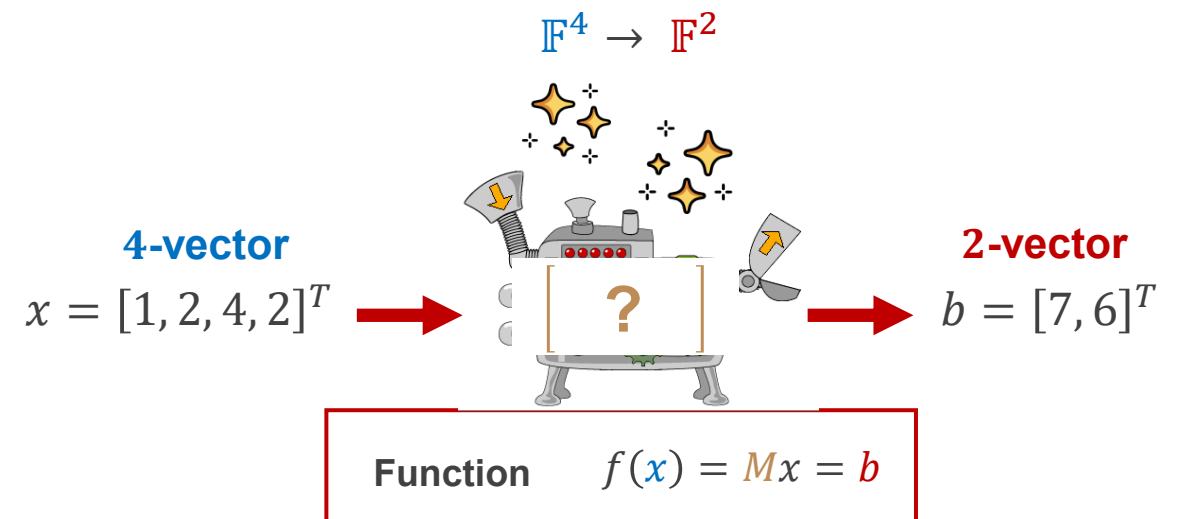


Matrix - Linear Functions (I)

We said e.g., we can (1) go from a matrix to a function (finding out input and output sizes) or (2) may find a matrix related to a set of inputs and outputs.



(1) Matrix to Function



(2) Function to Matrix

... but which functions can be expressed as a matrix-vector product?



Matrix - Linear Functions (II)

In the previous examples we simply assumed the **function** could be expressed as a matrix-vector product

How can we verify that assumption?

We'll state two algebraic properties.

- ▶ If a function can be expressed as a matrix-vector product $\mathbf{x} \mapsto M * \mathbf{x}$, it has these properties. **Function** as **matrix-vector product** $\rightarrow$ has **properties**
- ▶ If the function from $\mathbb{F}^C$ to $\mathbb{F}^R$ has these properties, it can be expressed as a matrix-vector product. **Function** has **properties** $\rightarrow$ as **matrix-vector product**



Matrix - Linear Functions (III)

Which functions can be expressed as a matrix-vector product?

Let $\mathcal{V}$ and $\mathcal{W}$ be vector spaces over a field $\mathbb{F}$.

Suppose a function $f : \mathcal{V} \longrightarrow \mathcal{W}$ satisfies two properties:

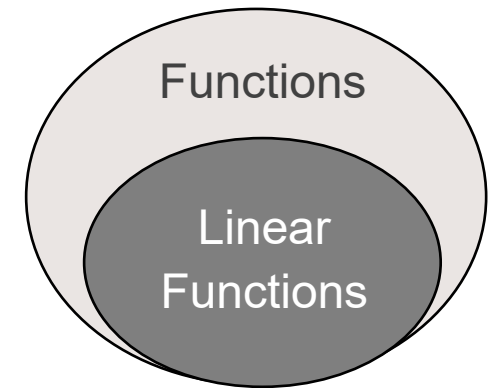
Property L1: For every vector $\mathbf{v}$ in $\mathcal{V}$ and every scalar α in $\mathbb{F}$,

$$f(\alpha \mathbf{v}) = \alpha f(\mathbf{v})$$

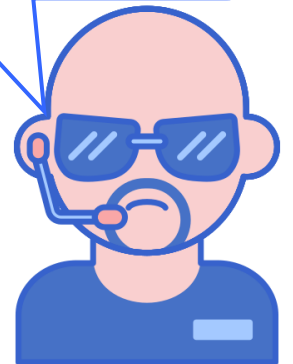
Property L2: For every two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathcal{V}$,

$$f(\mathbf{u} + \mathbf{v}) = f(\mathbf{u}) + f(\mathbf{v})$$

We then call f a linear function.



Sorry, not all
are accepted





Matrix - Linear Functions (IV)

Recap: Matrix-Vector Mult. - Algebraic Properties (from last week)

Proposition: Let A be an $R \times C$ matrix.

- For any C -vector $\mathbf{v}$ and any scalar α ,

$$A * (\alpha \mathbf{v}) = \alpha (A * \mathbf{v})$$

- For any C -vectors $\mathbf{u}$ and $\mathbf{v}$,

$$A * (\mathbf{u} + \mathbf{v}) = A * \mathbf{u} + A * \mathbf{v}$$



Matrix - Linear Functions (V)

Property L1: For every vector $\mathbf{v}$ in $\mathcal{V}$ and every scalar α in $\mathbb{F}$,

$$f(\alpha \mathbf{v}) = \alpha f(\mathbf{v})$$

Property L2: For every two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathcal{V}$,

$$f(\mathbf{u} + \mathbf{v}) = f(\mathbf{u}) + f(\mathbf{v})$$



Proposition: Let M be an $R \times C$ matrix, and suppose $f : \mathbb{F}^C \mapsto \mathbb{F}^R$ is defined by $f(\mathbf{x}) = M * \mathbf{x}$. Then f is a linear function.

Proof: Certainly $\mathbb{F}^C$ and $\mathbb{F}^R$ are vector spaces.

We showed that $M * (\alpha \mathbf{v}) = \alpha M * \mathbf{v}$. This proves that f satisfies Property L1.

We showed that $M * (\mathbf{u} + \mathbf{v}) = M * \mathbf{u} + M * \mathbf{v}$. This proves that f satisfies Property L2.

QED



Matrix - Linear Functions (VI)

Which functions are linear?

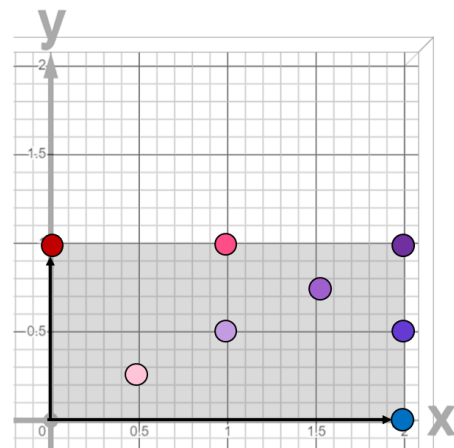
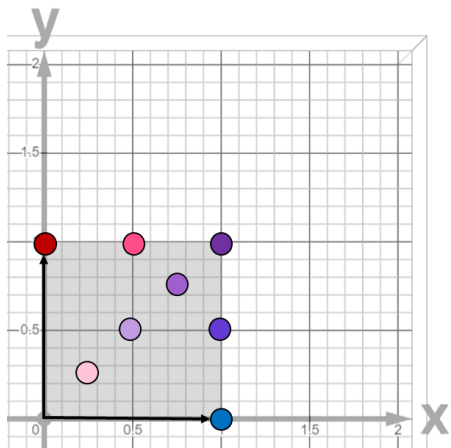
Stretching / Scaling

$$M = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

Property L1: $f(\alpha \mathbf{v}) = \alpha f(\mathbf{v})$

$$f\left(0.5 \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = f\left(\begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}\right) = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} * [0.5, 0.5]^T = 0.5 * \begin{bmatrix} 2 \\ 0 \end{bmatrix} + 0.5 * \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}$$

$$0.5 * f\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = 0.5 \left(\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} * [1, 1]^T \right) = 0.5 \left(1 * \begin{bmatrix} 2 \\ 0 \end{bmatrix} + 1 * \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = 0.5 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}$$



Property L2: $f(\mathbf{u} + \mathbf{v}) = f(\mathbf{u}) + f(\mathbf{v})$

$$f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = f\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = f\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$



Linear Function 



Matrix - Linear Functions (VII)

Which functions are linear?

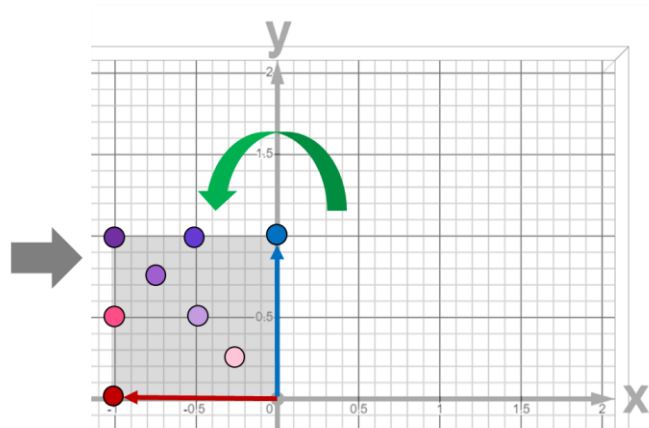
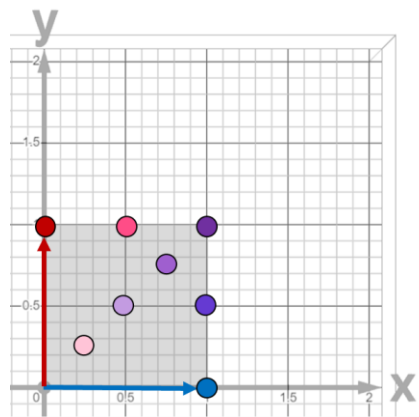
Property L1: $f(\alpha \mathbf{v}) = \alpha f(\mathbf{v})$

Rotation (e.g., 90 degrees)

$$M = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$f\left(0.5 \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = f\left(\begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}\right) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} * \begin{bmatrix} 0.5 & 0.5 \end{bmatrix}^T = 0.5 * \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 0.5 * \begin{bmatrix} -1 \\ 0 \end{bmatrix} = \begin{bmatrix} -0.5 \\ 0.5 \end{bmatrix}$$

$$0.5 * f\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = 0.5 \left(\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} * \begin{bmatrix} 1 & 1 \end{bmatrix}^T \right) = 0.5 \left(1 * \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 1 * \begin{bmatrix} -1 \\ 0 \end{bmatrix} \right) = 0.5 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -0.5 \\ 0.5 \end{bmatrix}$$



Property L2: $f(\mathbf{u} + \mathbf{v}) = f(\mathbf{u}) + f(\mathbf{v})$

$$f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = f\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) + f\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = f\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$



Linear Function





Matrix - Linear Functions (VIII)

Which functions are linear?

Translation

$$f\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Property L1: $f(\alpha \mathbf{v}) = \alpha f(\mathbf{v})$

$$f\left(0.5 \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = f\left(\begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}\right) = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1.5 \\ 2.5 \end{bmatrix}$$

$$0.5 * f\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = 0.5 \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \end{bmatrix}\right) = 0.5 \left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} 1.5 \\ 1.5 \end{bmatrix}$$



Property L2: $f(\mathbf{u} + \mathbf{v}) = f(\mathbf{u}) + f(\mathbf{v})$

$$f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) + f\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \end{bmatrix}\right) + \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

$$f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = f\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$



→ NOT a Linear Function 



Matrix - Linear Functions (IX)

Remember
(some slides
ago)

- ▶ If the function from $\mathbb{F}^C$ to $\mathbb{F}^R$ has these properties, it can be expressed as a matrix-vector product. **Function** has **properties** → as **matrix-vector product**

Stretching / Scaling



Linear Function



Can be expressed

$$x \mapsto M * x$$

Rotation



Linear Function



Can be expressed

$$x \mapsto M * x$$

Translation (adding a vector)



NOT a Linear Function



**Can NOT be
expressed $x \mapsto M * x$**

Since translation does not fulfill both properties, it cannot be expressed via $M * x$





Linear Functions & Linear Combination

Defining properties of linear functions:

Property L1: $f(\alpha \mathbf{v}) = \alpha f(\mathbf{v})$

Property L2: $f(\mathbf{u} + \mathbf{v}) = f(\mathbf{u}) + f(\mathbf{v})$

(We'll use this soon with standard generators for a proof)



Proposition: For a linear function f ,
for any vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ in the domain of f and any scalars $\alpha_1, \dots, \alpha_n$,

$$f(\alpha_1 \mathbf{v}_1 + \dots + \alpha_n \mathbf{v}_n) = \alpha_1 f(\mathbf{v}_1) + \dots + \alpha_n f(\mathbf{v}_n)$$

Proof: Consider the case of $n = 2$.

$$\begin{aligned} f(\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2) &= f(\alpha_1 \mathbf{v}_1) + f(\alpha_2 \mathbf{v}_2) \\ &= \alpha_1 f(\mathbf{v}_1) + \alpha_2 f(\mathbf{v}_2) \end{aligned}$$

by Property L2

by Property L1

Proof for general n is similar.

QED





From Function to Matrix - Revisited (I)

We saw a method to derive a matrix from a function:

Given a function $f : \mathbb{R}^n \longrightarrow \mathbb{R}^m$, we want a matrix M such that $f(\mathbf{x}) = M * \mathbf{x} \dots$

- ▶ Plug in the standard generators $\mathbf{e}_1 = [1, 0, \dots, 0, 0], \dots, \mathbf{e}_n = [0, \dots, 0, 1]$
- ▶ Column i of M is $f(\mathbf{e}_i)$.

This works correctly whenever such a matrix M really exists:





If it exists!



From Function to Matrix - Revisited (II)

Proof: If there is such a matrix then f is linear:

- ▶ (Property L1) $f(\alpha \mathbf{v}) = \alpha f(\mathbf{v})$ and
- ▶ (Property L2) $f(\mathbf{u} + \mathbf{v}) = f(\mathbf{u}) + f(\mathbf{v})$

Let $\mathbf{v} = [\alpha_1, \dots, \alpha_n]$ be any vector in $\mathbb{R}^n$.

We can write $\mathbf{v}$ in terms of the standard generators.

$$\mathbf{v} = \alpha_1 \mathbf{e}_1 + \dots + \alpha_n \mathbf{e}_n$$

so

$$\begin{aligned} f(\mathbf{v}) &= f(\alpha_1 \mathbf{e}_1 + \dots + \alpha_n \mathbf{e}_n) \\ &= \alpha_1 f(\mathbf{e}_1) + \dots + \alpha_n f(\mathbf{e}_n) \\ &= \alpha_1 (\text{column 1 of } M) + \dots + \alpha_n (\text{column } n \text{ of } M) \\ &= M * \mathbf{v} \quad \text{QED} \end{aligned}$$





Linear Function - Kernel

Remember
(last lecture)

Definition: Null space of a matrix A is $\{\mathbf{u} : A * \mathbf{u} = \mathbf{0}\}$. Written $\text{Null } A$

Definition: Kernel of a linear function f is $\{\mathbf{v} : f(\mathbf{v}) = \mathbf{0}\}$

Written $\text{Ker } f$

For a function $f(\mathbf{x}) = M * \mathbf{x}$,

$$\text{Ker } f = \text{Null } M$$



So, the kernel of f is the subset of the domain (inputs) that is mapped to the zero vector





Linear Function - Kernel and one-to-one (I)

One-to-One Lemma: A linear function is one-to-one if and only if its kernel is a trivial vector space.

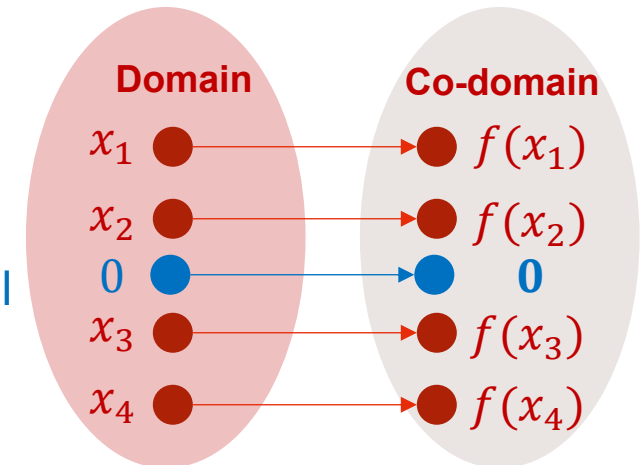
Definition: Kernel of a linear function f is $\{\mathbf{v} : f(\mathbf{v}) = \mathbf{0}\}$

Remember:

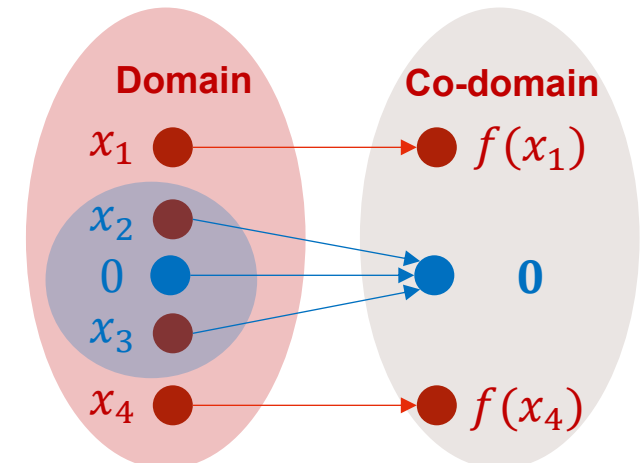
A **trivial vector space** is a vector space only consisting of the **zero vector**

Seems a bit obvious: if more than one input leads to the **zero vector**, then the function cannot be **one-to-one**

$\text{Ker } f$ is a trivial vector space



$\text{Ker } f$ is **NOT** a trivial vector space





Linear Function - Kernel and one-to-one (II)

One-to-One Lemma: A linear function is one-to-one if and only if its kernel is a trivial vector space. (= contains only the zero vector)

i.e., there is only one input that leads to the **zero vector** as output



Proof: Let $f : \mathcal{U} \longrightarrow \mathcal{V}$ be a linear function. We prove two directions.

- ▶ Suppose $\text{Ker } f$ contains some nonzero vector $\mathbf{u}$, so $f(\mathbf{u}) = \mathbf{0}_{\mathcal{V}}$. Because a linear function maps zero to zero, $f(\mathbf{0}) = \mathbf{0}_{\mathcal{V}}$ as well, so f is not one-to-one.
- ▶ Suppose $\text{Ker } f = \{\mathbf{0}\}$.
Let $\mathbf{v}_1, \mathbf{v}_2$ be any vectors such that $f(\mathbf{v}_1) = f(\mathbf{v}_2)$. (Two inputs lead to the same output)
Then $f(\mathbf{v}_1) - f(\mathbf{v}_2) = \mathbf{0}_{\mathcal{V}}$
so, by linearity, $f(\mathbf{v}_1 - \mathbf{v}_2) = \mathbf{0}_{\mathcal{V}}$,
so $\mathbf{v}_1 - \mathbf{v}_2 \in \text{Ker } f$.
Since $\text{Ker } f$ consists solely of $\mathbf{0}$,
it follows that $\mathbf{v}_1 - \mathbf{v}_2 = \mathbf{0}$, so $\mathbf{v}_1 = \mathbf{v}_2$. QED



Linear Function - Kernel and one-to-one (III)

One-to-One Lemma A linear function is one-to-one if and only if its kernel is a trivial vector space.

Define the function $f(\mathbf{x}) = A * \mathbf{x}$.

If $\text{Ker } f$ is trivial (i.e. if $\text{Null } A$ is trivial)

then a vector $\mathbf{b}$ is the image under f of at most one vector.

That is, at most one vector $\mathbf{u}$ such that $A * \mathbf{u} = \mathbf{b}$

That is, the solution set of $A * \mathbf{x} = \mathbf{b}$ has at most one vector.

(In simple terms:

If $\text{Ker } f$ is trivial $\rightarrow$ one-to-one $\rightarrow$ only one input $\mathbf{u}$ (solution) can get to the output $\mathbf{b}$)

UNIQUE SOLUTION





Linear Function - Kernel and onto (I)

Recall: for a function $f : \mathcal{V} \longrightarrow \mathcal{W}$, the *image* of f is the set of all images of elements of the domain:

$$\{f(\mathbf{v}) : \mathbf{v} \in \mathcal{V}\}$$

(You might know it as the “range” but we avoid that word here.)

The image of function f is written $\text{Im } f$

“Is function f onto?” same as “is $\text{Im } f = \text{co-domain of } f$?”

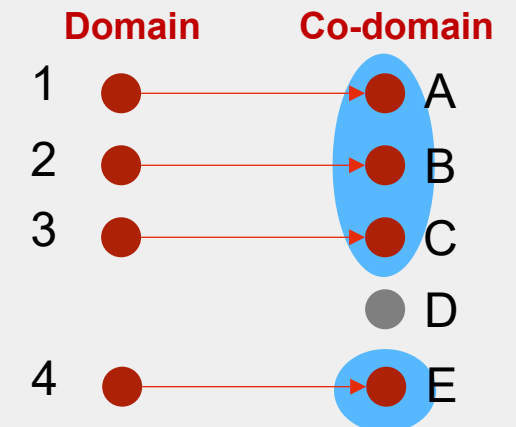
The image of f is not necessarily the co-domain of f

Remember (Lecture 1)

Example: a function $f : \{1, 2, 3, 4\} \longrightarrow \{'A', 'B', 'C', 'D', 'E'\}$

The image of f is $\text{Im } f = \{'A', 'B', 'C', 'E'\}$

'D' is in the co-domain but not in the image.





Linear Function - Kernel and onto (II)

We will not be able to give that answer yet, but we can study the image of linear function f a bit more ...

Proposition: The image of a linear function $f : \mathcal{V} \longrightarrow \mathcal{W}$ is a vector space

Welcome back old friend, **properties of a vector space (V1,V2, V3)**

Recall: a set $\mathcal{U}$ of vectors is a vector space if

V1: $\mathcal{U}$ contains a zero vector,

V2: for every vector $\mathbf{w}$ in $\mathcal{U}$ and every scalar α , the vector $\alpha \mathbf{w}$ is in $\mathcal{U}$

V3: for every pair of vectors $\mathbf{w}_1$ and $\mathbf{w}_2$ in $\mathcal{U}$, the vector $\mathbf{w}_1 + \mathbf{w}_2$ is in $\mathcal{U}$





Linear Function - Kernel and onto (III)



Proof:

V1: Since the domain $\mathcal{V}$ contains a zero vector $\mathbf{0}_{\mathcal{V}}$ and $f(\mathbf{0}_{\mathcal{V}}) = \mathbf{0}_{\mathcal{W}}$, the image of f includes $\mathbf{0}_{\mathcal{W}}$. This proves Property V1.

V2: Suppose some vector $\mathbf{w}$ is in the image of f .
That means there is some vector $\mathbf{v}$ in the domain $\mathcal{V}$ that maps to $\mathbf{w}$: $f(\mathbf{v}) = \mathbf{w}$.
By Property L1, for any scalar α , $f(\alpha \mathbf{v}) = \alpha f(\mathbf{v}) = \alpha \mathbf{w}$
so $\alpha \mathbf{w}$ is in the image. This proves Property V2.

V3: Suppose vectors $\mathbf{w}_1$ and $\mathbf{w}_2$ are in the image of f .
That is, there are vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ in the domain such that $f(\mathbf{v}_1) = \mathbf{w}_1$ and $f(\mathbf{v}_2) = \mathbf{w}_2$.
By Property L2, $f(\mathbf{v}_1 + \mathbf{v}_2) = f(\mathbf{v}_1) + f(\mathbf{v}_2) = \mathbf{w}_1 + \mathbf{w}_2$
so $\mathbf{w}_1 + \mathbf{w}_2$ is in the image. This proves Property V3.

QED



Linear Function - Kernel and onto (IV)

We've shown

Proposition: The image of a linear function $f : \mathcal{V} \longrightarrow \mathcal{W}$ is a vector space

This proposition shows that, for a linear function f , $\text{Im } f$ is always a subspace of the co-domain $\mathcal{W}$.

The function is onto if $\text{Im } f$ includes all of $\mathcal{W}$.
In a couple of weeks we will have a way to tell.





Matrix-Matrix Multiplication (I)

If

- ▶ A is a $R \times S$ matrix, and
- ▶ B is a $S \times T$ matrix

then it is legal to multiply A times B .

- ▶ In Mathese, written AB
- ▶ In our Mat class, written $A*B$

AB is different from BA .

In fact, one product might be legal while the other is illegal.

$$\begin{array}{c}
 \textcolor{red}{A} \qquad \qquad \textcolor{blue}{B} \\
 \left[\begin{array}{c} \end{array} \right] * \left[\begin{array}{c} \end{array} \right] = \left[\begin{array}{c} \end{array} \right] \\
 \textcolor{red}{R} \times \textcolor{red}{S} \quad \textcolor{green}{S} \times \textcolor{blue}{T} \qquad \qquad \textcolor{red}{R} \times \textcolor{blue}{T}
 \end{array}$$

$$\begin{array}{c}
 \textcolor{blue}{B} \qquad \qquad \textcolor{red}{A} \\
 \left[\begin{array}{c} \end{array} \right] * \left[\begin{array}{c} \end{array} \right] \rightarrow \text{Math Police} \\
 \textcolor{blue}{S} \times \textcolor{blue}{T} \quad \textcolor{red}{R} \times \textcolor{red}{S}
 \end{array}$$





Matrix-Matrix Multiplication via Dot Product

Combine

- ▶ *matrix-vector* definition of matrix-matrix multiplication, and
- ▶ *dot-product* definition of matrix-vector multiplication

to get...

Dot-product definition of matrix-matrix multiplication:

Entry rc of AB is the dot-product of row r of A with column c of B .

Example:

$$\begin{bmatrix} \boxed{1} & \boxed{0} & \boxed{2} \\ \boxed{3} & \boxed{1} & \boxed{0} \\ \boxed{2} & \boxed{0} & \boxed{1} \end{bmatrix} \begin{bmatrix} \boxed{2} & \boxed{1} \\ \boxed{5} & \boxed{0} \\ \boxed{1} & \boxed{3} \end{bmatrix} = \begin{bmatrix} \boxed{[1, 0, 2]} \cdot \boxed{[2, 5, 1]} & \boxed{[1, 0, 2]} \cdot \boxed{[1, 0, 3]} \\ \boxed{[3, 1, 0]} \cdot \boxed{[2, 5, 1]} & \boxed{[3, 1, 0]} \cdot \boxed{[1, 0, 3]} \\ \boxed{[2, 0, 1]} \cdot \boxed{[2, 5, 1]} & \boxed{[2, 0, 1]} \cdot \boxed{[1, 0, 3]} \end{bmatrix} = \begin{bmatrix} 4 & 7 \\ 11 & 3 \\ 5 & 5 \end{bmatrix}$$



Matrix-Matrix Multiplication - Transpose (I)

$$(AB)^T = B^T A^T$$

Example:

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 4 \\ 19 & 8 \end{bmatrix}$$

$$(AB)^T = \begin{bmatrix} 7 & 4 \\ 19 & 8 \end{bmatrix}^T = \begin{bmatrix} 7 & 19 \\ 4 & 8 \end{bmatrix}$$

$$B^T A^T = \begin{bmatrix} 5 & 0 \\ 1 & 2 \end{bmatrix}^T \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}^T = \begin{bmatrix} 5 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 7 & 19 \\ 4 & 8 \end{bmatrix}$$



Matrix-Matrix Multiplication - Transpose (II)



$$(AB)^T = A^T B^T \quad \text{✗}$$



$$(AB)^T = B^T A^T \quad \text{✓}$$

You might think “ $(AB)^T = A^T B^T$ ” but this is **false**.

In fact, doesn't even make sense!

- ▶ For AB to be legal, A 's column labels = B 's row labels.
- ▶ For $A^T B^T$ to be legal, A 's row labels = B 's column labels.

Example: $\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} 6 & 7 \\ 8 & 9 \end{bmatrix}$ is legal but $\begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix} \begin{bmatrix} 6 & 8 \\ 7 & 9 \end{bmatrix}$ is not.

$3 \times 2 \quad 2 \times 2$
 $2 \times 3 \quad 2 \times 2$

This is your second warning!



Math Police



Exercise: Matrix Multiplication

You are given the following matrices:

- 1) If possible, what are the **expected sizes** of the matrices resulting from the following matrix-matrix multiplications: AB , AC , CB , DE . Show work.

- 2) From 1) calculate all matrix multiplications that are possible
- 3) Using matrix A and B above show that $(AB)^T$ is NOT equal to $A^T B^T$ but rather equal to $B^T A^T$.
- 4) With 2 of the given matrices show that being able to multiply one matrix with the other does not mean that is possible in the reverse order

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \quad C = \begin{bmatrix} 2 & 3 \\ -2 & 2 \\ 3 & 1 \end{bmatrix}$$
$$D = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 2 & 1 \\ 2 & 1 & 1 \end{bmatrix} \quad E = \begin{bmatrix} -1/3 & 0 & 2/3 \\ -1/3 & 1 & -1/3 \\ 1 & -1 & 0 \end{bmatrix}$$



Solution

Exercise: Matrix Multiplication

- a) $AB: (2 \times 2) \times (2 \times 2) = (2 \times 2)$ (A's No columns = B's No. rows)
 $AC: (2 \times 2) \times (3 \times 2) \rightarrow$ Not possible
 $CB: (3 \times 2) \times (2 \times 2) = (3 \times 2)$ (C's No columns = B's No. rows)
 $DE: (3 \times 3) \times (3 \times 3) = (3 \times 3)$ (D's No columns = E's No. rows)

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \quad C = \begin{bmatrix} 2 & 3 \\ -2 & 2 \\ 3 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 2 & 1 \\ 2 & 1 & 1 \end{bmatrix} \quad E = \begin{bmatrix} -1/3 & 0 & 2/3 \\ -1/3 & 1 & -1/3 \\ 1 & -1 & 0 \end{bmatrix}$$

b) $AB = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} \times \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 2 * 1 + (-1) * 1 & 2 * 2 + (-1) * 3 \\ 1 * 1 + 3 * 1 & 1 * 2 + 3 * 3 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 4 & 11 \end{bmatrix}$

$$CB = \begin{bmatrix} 2 & 3 \\ -2 & 2 \\ 3 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 2 * 3 + 3 * 1 & 2 * 2 + 3 * 3 \\ -2 * 1 + 2 * 1 & -2 * 2 + 2 * 3 \\ 3 * 1 + 1 * 1 & 3 * 2 + 1 * 3 \end{bmatrix} = \begin{bmatrix} 9 & 13 \\ 0 & 2 \\ 7 & 9 \end{bmatrix}$$

$$DE = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 2 & 1 \\ 2 & 1 & 1 \end{bmatrix} \times \begin{bmatrix} -1/3 & 0 & 2/3 \\ -1/3 & 1 & -1/3 \\ 1 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 * (-1/3) + 2 * (-1/3) + 2 * 1 & 1 * 0 + 2 * 1 + 2 * (-1) & 1 * (2/3) + 2 * (-1/3) + 2 * 0 \\ 1 * (-1/3) + 2 * (-1/3) + 1 * 1 & 1 * 0 + 2 * 1 + 1 * (-1) & 1 * (2/3) + 2 * (-1/3) + 1 * 0 \\ 2 * (-1/3) + 1 * (-1/3) + 1 * 1 & 2 * 0 + 1 * 1 + 1 * (-1) & 2 * (2/3) + 1 * (-1/3) + 1 * 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



Solution

Exercise: Matrix Multiplication

$$c) \quad (AB)^T = \left(\begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} \times \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \right)^T = \left(\begin{bmatrix} 1 & 1 \\ 4 & 11 \end{bmatrix} \right)^T = \begin{bmatrix} 1 & 4 \\ 1 & 11 \end{bmatrix}$$

$$A^T B^T = \left(\begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} \right)^T \times \left(\begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \right)^T = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} \times \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 5 \\ 5 & 8 \end{bmatrix}$$

$$B^T A^T = \left(\begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \right)^T \times \left(\begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} \right)^T = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix} \times \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 1 & 11 \end{bmatrix}$$

The results show $(AB)^T \neq A^T B^T$ but $(AB)^T = B^T A^T$

$$d) \quad CA = \begin{bmatrix} 2 & 3 \\ -2 & 2 \\ 3 & 1 \end{bmatrix} \times \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 7 & 7 \\ -2 & 8 \\ 7 & 0 \end{bmatrix}$$

$AC = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} \times \begin{bmatrix} 2 & 3 \\ -2 & 2 \\ 3 & 1 \end{bmatrix}$ Not possible: $(2 \times 2) \times (3 \times 2)$ does not work as the column number of A does not correspond to the row number of C. This illustrates that if a matrix can be multiplied by another, the reverse is not necessarily allowed

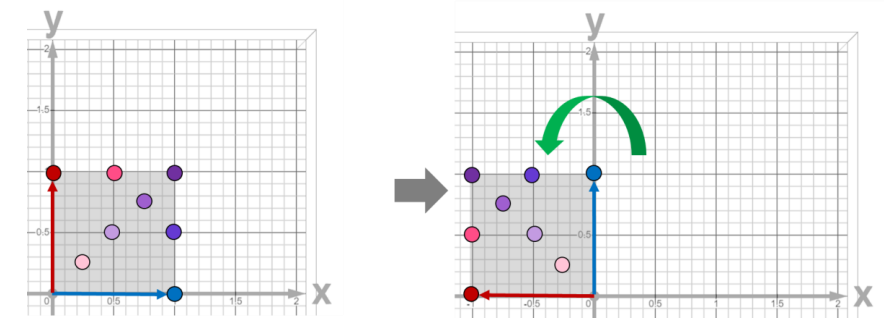
$$A = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \quad C = \begin{bmatrix} 2 & 3 \\ -2 & 2 \\ 3 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 2 & 1 \\ 2 & 1 & 1 \end{bmatrix} \quad E = \begin{bmatrix} -1/3 & 0 & 2/3 \\ -1/3 & 1 & -1/3 \\ 1 & -1 & 0 \end{bmatrix}$$



Exercise: Linear Functions

- 1) What is the matrix A for the linear function f that **stretches the horizontal direction (x) by a factor of three**? Show steps to determine the matrix
- 2) What is the matrix B for the linear function g that rotates points by **90 degrees** (see visualization)? Show steps to determine the matrix
- 3) Using matrix multiplication (order!), what is the matrix C for the linear function h that performs **both** – the stretching and the rotation? (Keep in mind that the order of operations matters, here: $B(Ax)$ - consider how C is calculated accordingly). Show steps to determine the matrix



rotation by 90 degrees



Solution

Exercise: Linear Functions

- 1) What is the matrix A for the linear function f that **stretches the horizontal direction (x) by a factor of three**? Show steps to determine the matrix
- 2) What is the matrix B for the linear function g that rotates points by **90 degrees** (see visualization)? Show steps to determine the matrix
- 3) Using matrix multiplication (order!), what is the matrix C for the linear function h that performs **both** – the stretching and the rotation? (Keep in mind that the order of operations matters, here: $B(Ax)$ - consider how C is calculated accordingly). Show steps to determine the matrix

$$1) \blacktriangleright \text{ We know } g \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

$$g \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\text{Therefore } B = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

$$2) \blacktriangleright \text{ We know } f \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$f \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$\text{Therefore } A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$3) C = BA = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} * \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 3 & 0 \end{bmatrix}$$

$$\text{This means: } f \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad f \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$\text{Therefore } C = \begin{bmatrix} 0 & -1 \\ 3 & 0 \end{bmatrix}$$



Some Key Take-aways (Lecture 9)

Standard generators can help understand what the effect of the multiplication with a matrix is (e.g., for scaling and rotation)

A **linear function** can be expressed as a matrix-vector multiplication and has two properties (L1, L2)

Link of the Matrix to
Linear Functions

Scaling and **rotation** are linear functions that can be expressed via matrix-vector multiplication, whereas **translation** (=vector addition) is not (and cannot be expressed via matrix-vector multiplication)

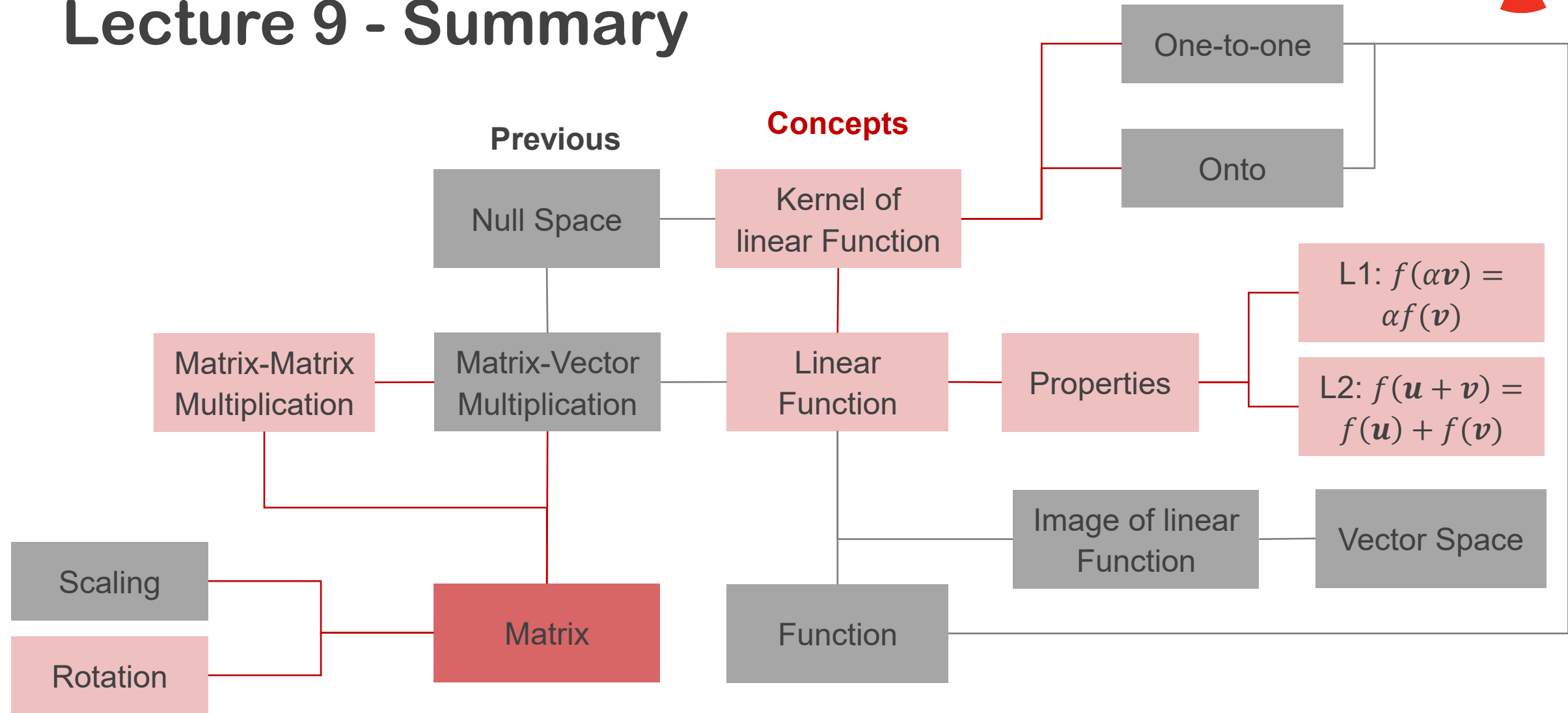
The **kernel of a linear function** is the set of inputs that are mapped by the function to the zero vector

Similar to Null
Space of a Matrix

Matrix-matrix multiplication can be implemented using the dot-product, using the dot product definition for matrix-vector multiplication



Lecture 9 - Summary





Thank You For Your Attention!



**Accomplishment unlocked:
Lecture 9 Completed**



APPENDIX



Linear Functions & Linear Combination (II)

Example: $f(\mathbf{x}) = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} * \mathbf{x}$

Remember (Previous slide)

$$f(\alpha_1 \mathbf{v}_1 + \cdots + \alpha_n \mathbf{v}_n) = \alpha_1 f(\mathbf{v}_1) + \cdots + \alpha_n f(\mathbf{v}_n)$$

Verify that $f(10[1, -1] + 20[1, 0]) = 10f([1, -1]) + 20f([1, 0])$

$$\begin{aligned} & \overset{M}{\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}} \left(\overset{\alpha_1}{10} \overset{v_1}{[1, -1]} + \overset{\alpha_2}{20} \overset{v_2}{[1, 0]} \right) \\ &= \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} ([10, -10] + [20, 0]) \\ &= \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} [30, -10] \\ &= 30[1, 3] - 10[2, 4] \\ &= [30, 90] - [20, 40] \\ &= [10, 50] \end{aligned}$$

$$\begin{aligned} & \overset{\alpha_1}{10} \left(\overset{M}{\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}} * \overset{v_1}{[1, -1]} \right) + \overset{\alpha_2}{20} \left(\overset{M}{\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}} * \overset{v_2}{[1, 0]} \right) \\ &= 10([1, 3] - [2, 4]) + 20(1[1, 3]) \\ &= 10[-1, -1] + 20[1, 3] \\ &= [-10, -10] + [20, 60] \\ &= [10, 50] \end{aligned}$$



Matrix-Matrix Multiplication (II)

We'll see two equivalent definitions:

- ▶ one in terms of **vector-matrix** multiplication,
- ▶ one in terms of **matrix-vector** multiplication.





Vector-Matrix Definition (I)

Vector-matrix definition of matrix-matrix multiplication:
For each row-label r of A ,

$$\text{row } r \text{ of } AB = \underbrace{(\text{row } r \text{ of } A)}_{\text{vector}} * B$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} B \end{bmatrix} = \begin{bmatrix} [1, 0, 0] * B \\ [2, 1, 0] * B \\ [0, 0, 1] * B \end{bmatrix}$$

How to interpret $[1, 0, 0] * B$? ←

- ▶ Linear combinations definition of vector-matrix multiplication?
- ▶ Dot-product definition of vector-matrix multiplication?

Each is correct.



Vector-Matrix Definition (II)

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} B \end{bmatrix} = \begin{bmatrix} [1, 0, 0] * B \\ [2, 1, 0] * B \\ [0, 0, 1] * B \end{bmatrix}$$

How to interpret $[1, 0, 0] * B$? Linear combinations definition:

$$\begin{array}{ll} \textcircled{1} \quad [1, 0, 0] * \begin{bmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \\ \mathbf{b}_3 \end{bmatrix} = \mathbf{b}_1 & \textcircled{3} \quad [0, 0, 1] * \begin{bmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \\ \mathbf{b}_3 \end{bmatrix} = \mathbf{b}_3 \\ \textcircled{2} \quad [2, 1, 0] * \begin{bmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \\ \mathbf{b}_3 \end{bmatrix} = 2\mathbf{b}_1 + \mathbf{b}_2 \end{array}$$

Conclusion:

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \\ \mathbf{b}_3 \end{bmatrix} = \begin{bmatrix} \mathbf{b}_1 \\ 2\mathbf{b}_1 + \mathbf{b}_2 \\ \mathbf{b}_3 \end{bmatrix} \begin{array}{l} \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \end{array}$$



Sparse Matrix-Vector Product (I)

Remember (Lecture 2)

A vector most of whose values are zero is called a *sparse* vector.

To compute matrix-vector or vector-matrix product,

- ▶ could use dot-product or linear-combinations definition.
- ▶ However, using those definitions, it's not easy to exploit sparsity in the matrix.

“Ordinary” Definition of Matrix-Vector Multiplication: If M is an $R \times C$ matrix and $\mathbf{u}$ is a C -vector then $M * \mathbf{u}$ is the R -vector $\mathbf{v}$ such that, for each $r \in R$,

$$v[r] = \sum_{c \in C} M[r, c] u[c]$$



Sparse Matrix-Vector Product (II)

“Ordinary” Definition of Matrix-Vector Multiplication: If M is an $R \times C$ matrix and $\mathbf{u}$ is a C -vector then $M * \mathbf{u}$ is the R -vector $\mathbf{v}$ such that, for each $r \in R$,

$$v[r] = \sum_{c \in C} M[r, c] u[c]$$

Possible Implementation:

1 for i in R :
 2 $v[i] := \sum_{j \in C} M[i, j] u[j]$

But this doesn't exploit sparsity!

Example:

$$\begin{bmatrix} 1 & 3 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} * [1 \quad 2 \quad 3] = \left[[1 * 1 + 3 * 2 + 0 * 3], [0 * 1 + 0 * 2 + 2 * 3], [0 * 1 + 0 * 2 + 0 * 3] \right]$$

$$= [7, 6, 0]$$

We use the dot product with all elements of the matrix



Sparse Matrix-Vector Product (III)

Idea:

- ▶ Initialize output vector $\mathbf{v}$ to zero vector.
- ▶ Iterate over nonzero entries of M , adding terms according to ordinary definition.

```

1 initialize  $\mathbf{v}$  to zero vector
2 for each pair  $(i,j)$  in sparse representation,
3    $v[i] = v[i] + M[i,j]u[j]$ 
  
```

Same Example:

$$\begin{bmatrix} 1 & 3 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} * \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 * 1 + 3 * 2, & 2 * 3, & 0 \end{bmatrix} \\
 = \begin{bmatrix} 7, 6, 0 \end{bmatrix}$$

Only consider entries of M that are non-zero and add dot products to a zero-vector



Matrix - Linear Functions (VI)

Lemma: If $f : \mathcal{U} \longrightarrow \mathcal{V}$ is a linear function then f maps the zero vector of $\mathcal{U}$ to the zero vector of $\mathcal{V}$.



Proof: Let $\mathbf{0}$ denote the zero vector of $\mathcal{U}$, and let $\mathbf{0}_{\mathcal{V}}$ denote the zero vector of $\mathcal{V}$.

$$f(\mathbf{0}) = f(\mathbf{0} + \mathbf{0}) = f(\mathbf{0}) + f(\mathbf{0})$$

Subtracting $f(\mathbf{0})$ from both sides, we obtain

$$\mathbf{0}_{\mathcal{V}} = f(\mathbf{0})$$

QED



Matrix-Vector Definition

Matrix-vector definition of matrix-matrix multiplication:

For each column-label s of B ,

$$\text{column } s \text{ of } AB = A * (\text{column } s \text{ of } B)$$

Let $A = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}$ and $B = \text{matrix with columns } [4, 3], [2, 1], \text{ and } [0, -1]$

$$B = \left[\begin{array}{c|c|c} 4 & 2 & 0 \\ 3 & 1 & -1 \end{array} \right]$$

AB is the matrix with column $i = A * (\text{column } i \text{ of } B)$

$$\textcircled{1} \quad A * [4, 3] = [10, -1]$$

$$\textcircled{2} \quad A * [2, 1] = [4, -1]$$

$$\textcircled{3} \quad A * [0, -1] = [-2, -1]$$

$$AB = \left[\begin{array}{c|c|c} 10 & 4 & -2 \\ -1 & -1 & -1 \end{array} \right] \begin{matrix} \textcircled{1} & \textcircled{2} & \textcircled{3} \end{matrix}$$